

3(a)	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1
3(b)(i)	1. $T \sin 53^\circ = 2.4$ $T = 3.0 \text{ N}$	A1
	2. $F = T \cos 53^\circ$ or $F^2 = T^2 - 2.4^2$ $F = 1.8 \text{ N}$	A1
3(b)(ii)	$\sigma = T/A$ or $\sigma = F/A$	C1
	$A = \pi d^2/4$ or $A = \pi r^2$	C1
	$\sigma = 3.0 \times 4 / [\pi \times (0.50 \times 10^{-3})^2]$ $= 1.5 \times 10^7 \text{ Pa}$	A1
3(c)(i)	$h = 75 - 75 \sin 53^\circ = 15 \text{ cm}$	A1
3(c)(ii)	$(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$	C1
	$(\Delta)E = 2.4 \times 15 \times 10^{-2}$ $= 0.36 \text{ J}$	A1
3(c)(iii)	$E = \frac{1}{2}mv^2$	B1
	$0.36 = \frac{1}{2} \times (2.4/9.81) \times v^2$	C1
	$v = 1.7 \text{ m s}^{-1}$	A1